Introduction to Mathematics and Optimization

May, 2023

PROBLEM 1

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function

$$f(x,y) = x^4 + y^4 + xy + 1$$

and $U \subseteq \mathbb{R}^2$ the subset

$$U = \left\{(x,y) \middle|\, x > \tfrac{1}{3},\, y > \tfrac{1}{3}\right\}.$$

(a) Show that the complement $\mathbb{R}^2 \setminus U$ is a closed subset of $\mathbb{R}^2$ and that U is an open subset of $\mathbb{R}^2$.

(b) Show that $144\, x^2 y^2 - 1 > 0$ if $(x,y) \in U$.

(c) Compute the Hessian matrix for f and conclude that $f : U \to \mathbb{R}$ is a convex function. It may be used without proof that U is a convex subset of $\mathbb{R}^2$.

(d) Show that $(0,0)$ is a critical point for $f : \mathbb{R}^2 \to \mathbb{R}$ and that it is neither a local maximum nor a local minimum.

(e) Find all critical points $\neq (0,0)$ for $f : \mathbb{R}^2 \to \mathbb{R}$ and show that they are local minima.

(f) Show that

$$f(x,y) \geq \tfrac{7}{8}$$

for all $(x,y) \in \mathbb{R}^2$ (hint: You may use the substitution $x = r\cos(t)$ and $y = r\sin(t)$. It may be used without proof that $\cos(t)^4 + \sin(t)^4 \geq \frac{1}{2}$ for $t \in \mathbb{R}$).

PROBLEM 2

Consider (again) the function $f : \mathbb{R}^2 \to \mathbb{R}$

$$f(x, y) = x^4 + y^4 + xy + 1$$

and the open convex subset

$$U = \left\{(x, y) \middle|\, x > \tfrac{1}{3}, y > \tfrac{1}{3}\right\}.$$

It is permitted to replace $\mathbb{R}^2$ with U when using results from the notes on convex optimization. From the previous problem, it is known that $f : U \to \mathbb{R}$ is a convex function. Let

$$F = \left\{(x, y) \middle|\, x \geq \tfrac{1}{2}, y \geq \tfrac{1}{2}\right\}.$$

(a) Why does $F \subseteq U$ hold?

(b) Is $f : \mathbb{R}^2 \to \mathbb{R}$ a convex function?

(c) Show that F is a closed convex subset. Is F a bounded subset?

(d) Write down the KKT conditions for the convex optimization problem

$$\begin{aligned} &\min f(v) \\ &v \in F \end{aligned} \tag{1}$$

(e) Use the KKT conditions for (1) to show that $x = \frac{1}{2}$ and $y = \frac{1}{2}$ is a minimum.

(f) How many solutions do the KKT conditions for (1) have?

PROBLEM 3

Let A_t denote the 2×2 matrix

$$A_t = \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$$

(a) Show that A_t is positive definite if and only if $-1 < t$ and $t < 1$.

(b) Show that A_t is positive semidefinite for $t \in \{-1, 1\}$. Here

$$A_{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \qquad \text{and} \qquad A_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

(c) Show that A_1 is not invertible.

(d) Show that

$$A_t^{-1} = \begin{pmatrix} 1/(1-t^2) & -t/(1-t^2) \\ -t/(1-t^2) & 1/(1-t^2) \end{pmatrix}.$$

for $t \notin \{-1, 1\}$.